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 Studierichting: Informatica
 Oefeningenreeks 7B nr. 7.2

$$\begin{aligned}
 Df(x, y) &= \begin{pmatrix} ye^{xy} & xe^{xy} \\ \frac{1}{x}D\left(\frac{x}{y}\right) & \frac{1}{x}D\left(\frac{x}{y}\right) \end{pmatrix} \\
 &= \begin{pmatrix} ye^{xy} & xe^{xy} \\ \frac{y}{x} \cdot \frac{1}{y}D(x) & \frac{y}{x}xD\left(\frac{1}{y}\right) \end{pmatrix} \\
 &= \begin{pmatrix} ye^{xy} & xe^{xy} \\ \frac{1}{x} & -\frac{1}{y} \end{pmatrix} \\
 -ye^{xy} \cdot \frac{1}{y} - \frac{1}{x} \cdot xe^{xy} &= -e^{xy} - e^{xy} = -2e^{xy} \neq 0 \Rightarrow \text{inverteerbaar}
 \end{aligned}$$

References